

EARTH SCIENCE & PHYSICAL GEOGRAPHY NOTES

Shaping of the Earth's Surface

Topics Covered: Plate Tectonics • Earth's Structure • Weathering & Erosion • Agents of Gradation • Exercise Solutions

CORE CONCEPT: LANDFORM

A **landform** is a natural feature on the Earth's surface formed by processes such as weathering, erosion, deposition, and the movement of the Earth's crust. Examples include mountains, valleys, plateaus, plains, deserts, and coastal features.

1. Plate Tectonics

Plate tectonics is a fundamental theory given by **W.J. Morgan** in earth science that explains the movement of the Earth's crust. According to this theory, the outermost layer of the Earth is broken into several large and small pieces called **tectonic plates**.

- Move slowly over the semi-molten layer beneath them (usually a few centimetres per year).
- Are responsible for major physical features and natural phenomena such as mountains, earthquakes, and volcanoes.

Structure of the Earth

The Earth is composed of three primary layers, with a total radius of approximately **6,375 km**:

Layer	Description & Characteristics
Crust	The outermost layer on which we live. Thickness: 30–40 km under continents; 5–7 km under oceans.
Mantle	The mostly solid, very thick and hot layer between the crust and outer core (approx. 2,900 km thick).
Core	The innermost layer, extremely hot and dense.

Sub-Layers & Key Terms:

- **Lithosphere:** The crust along with the rigid upper part of the mantle; broken into tectonic plates.
- **Asthenosphere:** The semi-molten, mobile layer beneath the lithosphere that allows plates to slide.

Core Components:

- **Outer Core:** A fluid layer mainly consisting of iron and nickel (approx. 2,200 km thick).
- **Inner Core:** A solid, hot spinning metal ball; the densest part of Earth (approx. 1,250 km thick).

Types of Tectonic Plates

- **Continental plates:** Carry continents.
- **Oceanic plates:** Carry ocean floors.
- **Mixed plates:** Carry both continents and oceans.

Major Tectonic Plates: Pacific, Eurasian, African, North American, South American, Indo-Australian, and Antarctic Plates.

Plate Dynamics & Boundaries

What Causes Plate Movement? Convection currents in the mantle drive plate movement. Heat from the core causes molten material to rise, while cooler material sinks. This continuous cycle pushes and pulls the tectonic plates.

Boundary Type	Movement & Process	Resulting Features / Examples
Convergent Boundary	Plates move towards each other. • <i>Continental + Continental</i> : Collide & fold. • <i>Oceanic + Continental</i> : Oceanic sinks (subducts).	• Forms fold mountains (e.g., Himalayas). • Volcanic activity, deep trenches & earthquakes.
Divergent Boundary	Plates move away from each other; magma rises to create new crust.	Mid-ocean ridges (e.g., Mid-Atlantic Ridge).
Transform Boundary	Plates slide horizontally past each other; neither creates nor destroys crust.	Severe earthquakes (e.g., San Andreas Fault, USA).

IMPORTANCE OF PLATE TECTONICS

Explains mountain building, ocean basin formation, volcanic belts, and seismic distribution. Most earthquakes and volcanoes occur along plate boundaries, particularly surrounding the Pacific Ocean—a region known as the **Ring of Fire**.

Ancient Indian Knowledge — Earthquakes (Bhūkampa)

In ancient India, earthquakes were called '*Bhūkampa*' (shaking of the Earth). Varāhamihira's *Bṛhatsaṃhitā* dedicated a special section to earthquakes, observing that changes in wind, rain, clouds, animal behavior, and planetary alignments could signal seismic events. He attributed earthquakes to four elemental forces: **Vāyu** (wind), **Agni** (fire), **Indra** (thunder), and **Varuṇa** (water).

2. Process of Weathering and Erosion

Weathering and erosion continuously break down and reshape the Earth's surface over long periods—wearing down mountains, carving valleys, forming plains, and creating caves, cliffs, and river deltas.

Weathering

The in-situ breaking down of rocks at the Earth's surface. *It does not involve the movement of material.*

Type	Mechanism	Example
Physical Weathering	Temperature changes, frost action, or wind breaking rocks into smaller fragments without chemical change.	Rocks splitting due to repeated expansion/contraction (thermal stress).
Chemical Weathering	Minerals in rocks alter due to chemical reactions with water, oxygen, or acids, creating new minerals.	Rusting of iron-rich rocks; dissolution of limestone in acidic water.
Biological Weathering	Disintegration caused by living organisms (plants, animals, microorganisms).	Plant roots expanding inside rock cracks and splitting them apart.

Erosion

The process by which soil and rock debris are worn away and **transported** from one place to another by natural agents.

Key Difference: *Weathering* only breaks down rocks in place; *Erosion* involves the active movement/transportation of broken materials.

Erosion Type	Primary Agent	Description
Water Erosion	Rivers, rain, ocean waves	Most common form of erosion worldwide.
Wind Erosion	Wind	Highly active in arid, dry, and sandy regions.
Glacial Erosion	Moving ice (glaciers)	Scrapes, scours, and carries bedrock debris.
Coastal Erosion	Sea waves & currents	Wears away landmasses along coastline margins.

Human Impact of Erosion

- **Agriculture:** Removes topsoil rich in nutrients, diminishing crop yield.
- **Infrastructure:** Washes away riverside roads, bridges, and coastal settlements.
- **Safety:** Destabilizes slopes during construction/mining, increasing landslide hazards.

Ancient Indian Soil and Water Conservation

The **Indus-Sarasvati Civilization** employed sophisticated techniques including contouring, bunding, terracing, dams, and canals. Classical Sanskrit texts like the *Vedas*, *Kṛṣiparāśara*, *Kauṭilya's Arthaśāstra*, and *Kṛṣṇiyurveda* document advanced land fertility assessment and water management.

The traditional **Zabo System** of Nagaland uses earthen bunds on hillslopes and check dams across streams to retard runoff, curb erosion, and deposit fertile silt.

- **Contouring (CCT):** Digging trenches along hillslope contour lines to trap water and recharge groundwater.
- **Bunding:** Earthen embankments built along contours to slow runoff.
- **Terracing:** Step-like level platforms built on steep slopes to prevent soil erosion.

3. Agents of Gradation & Landforms

Agents of gradation wear down high areas (degradation) and deposit sediments in low areas (aggradation), leveling the surface.

A. Running Water (Rivers)

Rivers shape landscapes across three distinct stages:

Course	Characteristics	Key Landforms Formed
Upper Course	Steep gradient, high velocity, dominant vertical erosion.	V-shaped valleys, waterfalls, rapids, gorges.
Middle Course	Gentler slope, river loses energy, lateral erosion dominant.	Meanders, oxbow lakes, floodplains.
Lower Course	Flat terrain, slow speed, heavy deposition.	Deltas, natural levees, alluvial fans.

Waterfall: River falls over a vertical cliff where hard rock overlies soft rock. Soft rock erodes into a plunge pool. Used for hydroelectric power & tourism.

Delta: Triangular landform at the river mouth built by deposition. Highly fertile (e.g., Ganga-Brahmaputra Delta).

B. Waves and Coastal Features

- **Beaches:** Depositional coastal landforms of sand/pebbles created by wave action. Act as natural storm barriers.
- **Erosional Features:** Cliffs (steep rock faces), Wave-cut platforms, Caves, Arches, and Stacks (isolated rock pillars).

C. Glaciers (Moving Ice)

Erosional Features: U-shaped valleys, Cirques (bowl-shaped hollows), Arêtes (sharp ridges), Hanging valleys, and Fjords (flooded glacial troughs).

Moraines (Depositional): Accumulations of glacial debris (till). Classified as *Lateral* (sides), *Terminal* (snout/end), or *Medial* (center junction).

D. Wind (Eolian Processes)

Erosional Landforms: Yardangs (streamlined ridges), Ventifacts (sandblasted stones), Deflation hollows, and Desert pavements.

Sand Dunes: Barchan (crescent-shaped, single-direction wind), Longitudinal (parallel ridges), Star dunes (multidirectional winds), Parabolic (U-shaped, vegetated).

E. Underground Water (Karst Topography)

Forms in soluble rocks like limestone via chemical dissolution.

- **Caves & Sinkholes (Dolines):** Subterranean hollows and surface collapses.
- **Speleothems:** Cave deposits created by mineral-rich drips.

 **Memory Tip:** *Stalactites* hang **tight** to the ceiling; *Stalagmites* **might** reach up from the floor!

4. Exercise Questions & Solutions

Q1. What are the sources of energy that cause internal (endogenic) Earth movements?

Endogenic movements are driven by internal heat energy generated through three main sources:

1. **Radioactive Decay:** Breakdown of radioactive isotopes (Uranium, Thorium, Potassium) in the interior releases immense thermal energy.
2. **Residual Heat:** Primordial heat trapped since the Earth's original accretion billions of years ago.
3. **Gravitational Compression:** Pressure from overburden layers generating deep thermal energy.

This heat drives mantle convection currents, propelling tectonic plates.

Q2. Relate India's major physiographic divisions with the endogenic forces responsible for their origin.

Physiographic Division	Responsible Endogenic Force / Process
Himalayan Mountains	Convergent collision between Indo-Australian and Eurasian plates (folding).
Northern Plains	Deposition of alluvium in the foreland basin created by the Himalayan uplift.
Peninsular Plateau	Ancient continental block derived from Gondwanaland breakup; ancient volcanic floods.
Western & Eastern Ghats	Faulting along continental margins combined with prolonged denudation.
Islands (Andaman & Nicobar)	Volcanic and subduction activity along the active plate margin (Ring of Fire extension).

Q3. Why and where do earthquakes occur frequently? Is exact prediction possible?

Why: Earthquakes occur when stress built up along tectonic faults exceeds friction, causing sudden slippage and releasing energy as seismic waves.

Where: Concentrated along plate boundaries: Convergent zones (Himalayas), Transform faults (San Andreas), and subduction rims (Pacific Ring of Fire—accounting for ~90% of global quakes).

Prediction: Precise short-term prediction (exact date, time, location) is currently *not possible*. However, scientists construct seismic hazard maps and monitor foreshocks, ground tilt, and strain levels for risk management.

Q4. "Plate movements are responsible for the distribution of earthquakes and volcanoes." Explain.

Plate boundary interactions directly control igneous and seismic activity:

- **Subduction / Convergent Zones:** Sinking oceanic plates melt under high temperature/pressure, generating ascending magma (volcanism) and deep-focus earthquakes.
- **Divergent Zones:** Separating plates allow upwelling mantle material to form mid-oceanic ridges and basaltic fissure eruptions.
- **Transform Margins:** Lateral grinding creates shallow, high-intensity earthquakes without major volcanism.

Q5. How are deforestation and soil erosion directly associated?

Vegetation provides critical protection against erosion:

- **Root Systems:** Bind soil aggregates together structurally.
- **Canopy Cover:** Absorbs kinetic energy of raindrops, preventing splash erosion.
- **Hydrological Regulation:** Promotes infiltration and retards surface runoff velocity.

Removing tree cover exposes bare soil directly to splash, sheet, and gully erosion, resulting in severe topsoil depletion and hillside instability.

Q6. What precautionary measures should be taken in an earthquake-prone zone?

- **Before:** Structural retrofitting, securing heavy furniture, preparing an emergency survival kit, and practicing *Drop, Cover, and Hold On* drills.
- **During:** Take cover under sturdy furniture away from glass/exterior walls; if outdoors, move to open ground away from poles/buildings.
- **After:** Inspect for gas leaks/structural damage, avoid elevators, stay updated via emergency broadcasts, and expect aftershocks.